

BEV Risk Map Analysis Report

Occlusion-Based Emergence Risk Assessment

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Tool: BEV-RiskViz (BEV Risk Visualization Tool)

Purpose: Autonomous driving risk assessment based on occlusion analysis

Scenario Information:

Scenario: Multi-Vehicle Intersection

Risk Calculation Parameters:

weight_trajectory: 0.3

weight_occlusion: 0.3

weight_temporal: 0.2

weight_proximity: 0.2

bev_x_range: (-50.0, 50.0)

bev_y_range: (-50.0, 50.0)

bev_resolution: 0.5

ego_velocity: 10.0

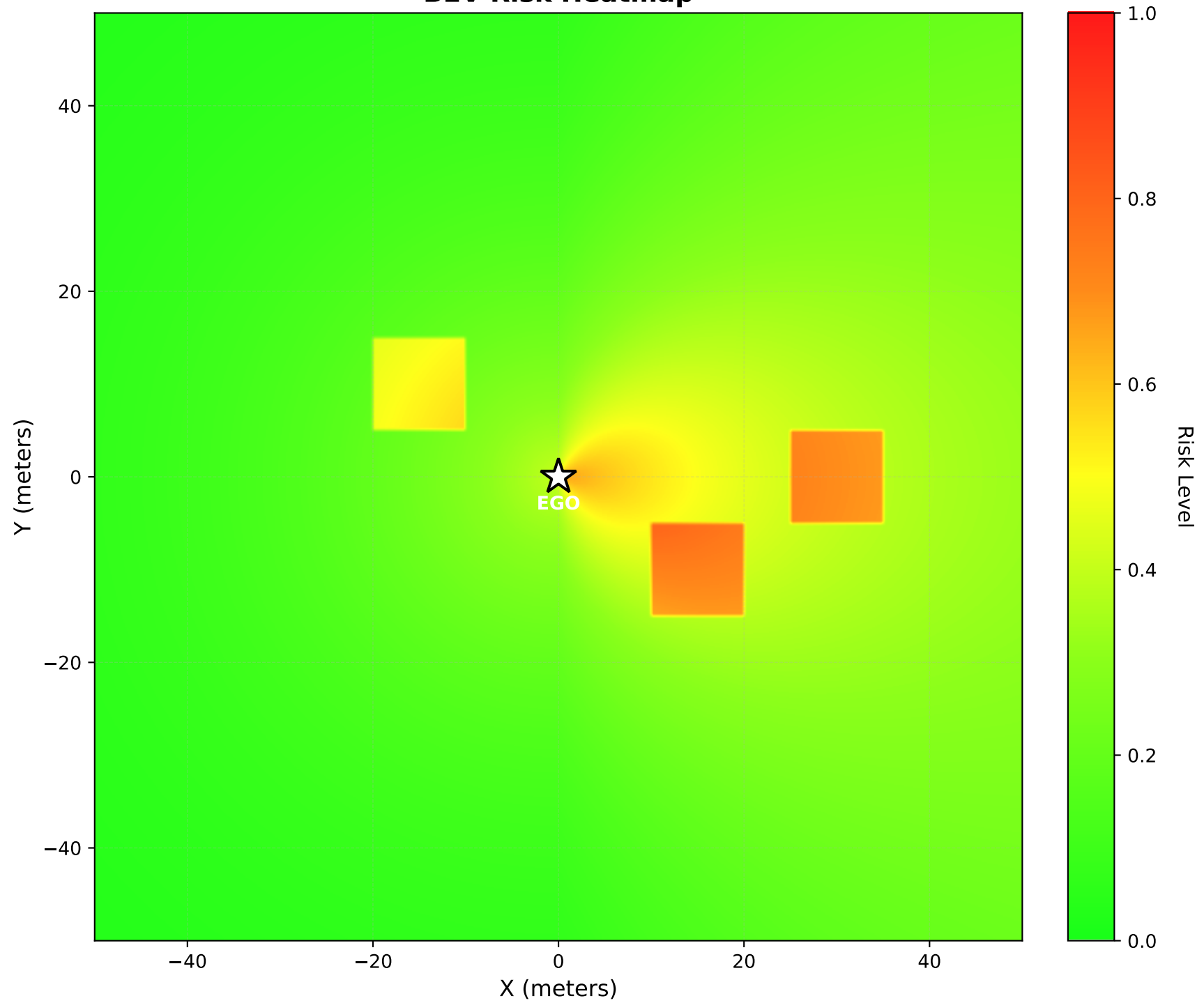
ego_heading: 0.0

max_proximity_distance: 50.0

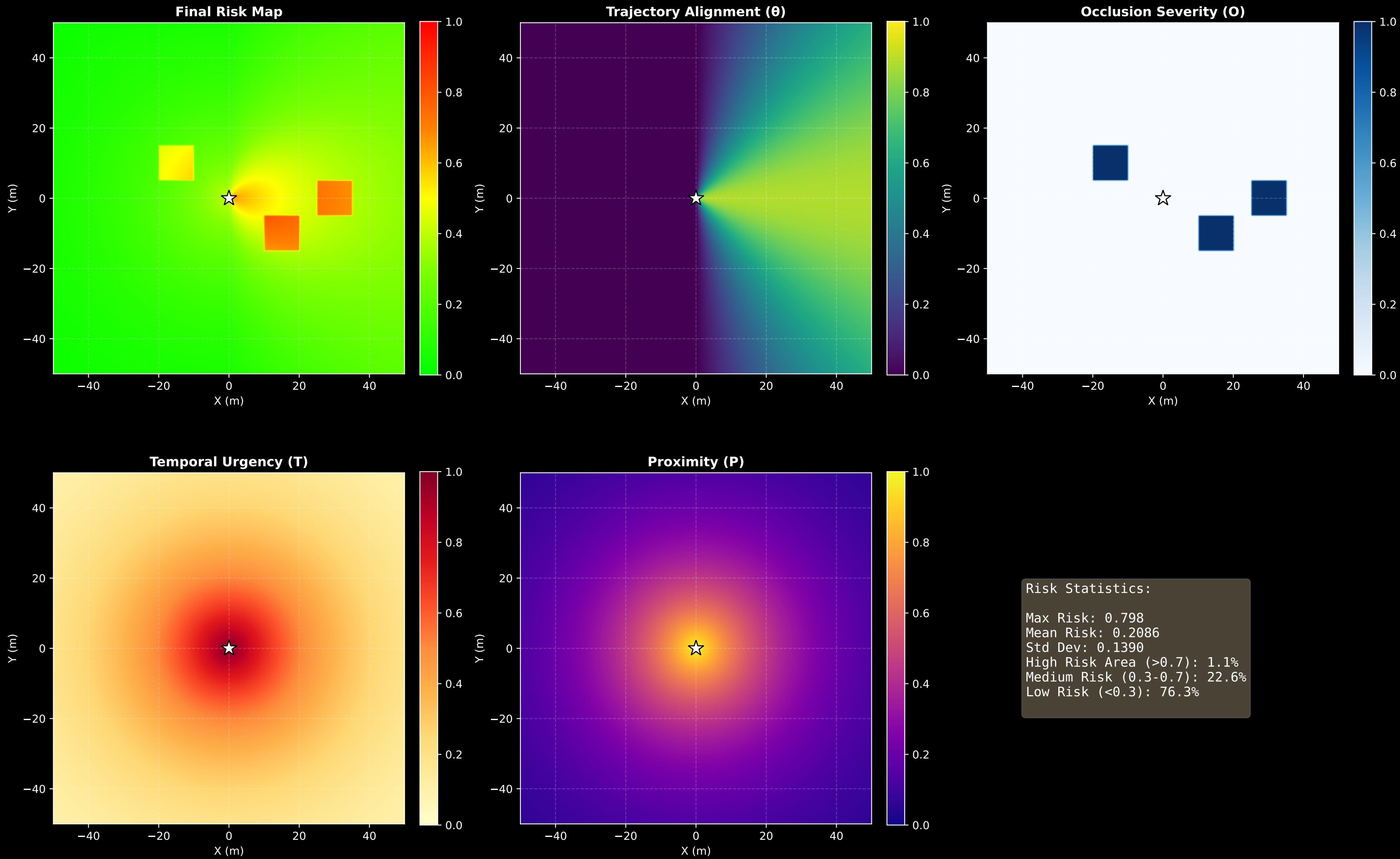
time_horizon: 3.0

occlusion_depth_threshold: 2.0

BEV Risk Heatmap



Risk Factor Analysis



Statistical Analysis

OVERALL RISK STATISTICS

Risk Map Dimensions: 200 × 200 cells
Total Cells: 40,000

Risk Value Statistics:

Maximum Risk: 0.797602
Minimum Risk: 0.031154
Mean Risk: 0.208628
Median Risk: 0.183584
Std Deviation: 0.138956

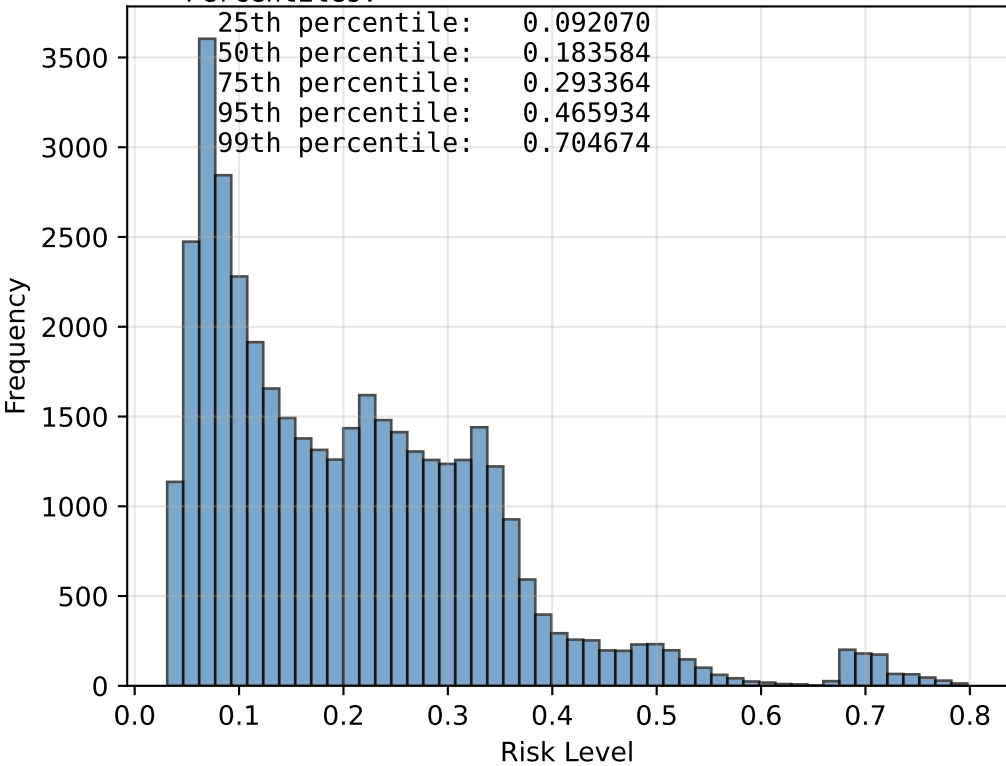
Risk Distribution:

High Risk (>0.7): 454 cells (1.14%)
Medium Risk (0.3-0.7): 9,028 cells (22.57%)
Low Risk (<0.3): 30,518 cells (76.30%)

Risk Distribution Histogram

Percentiles:

25th percentile: 0.092070
50th percentile: 0.183584
75th percentile: 0.293364
95th percentile: 0.465934
99th percentile: 0.704674



Risk Category Distribution

